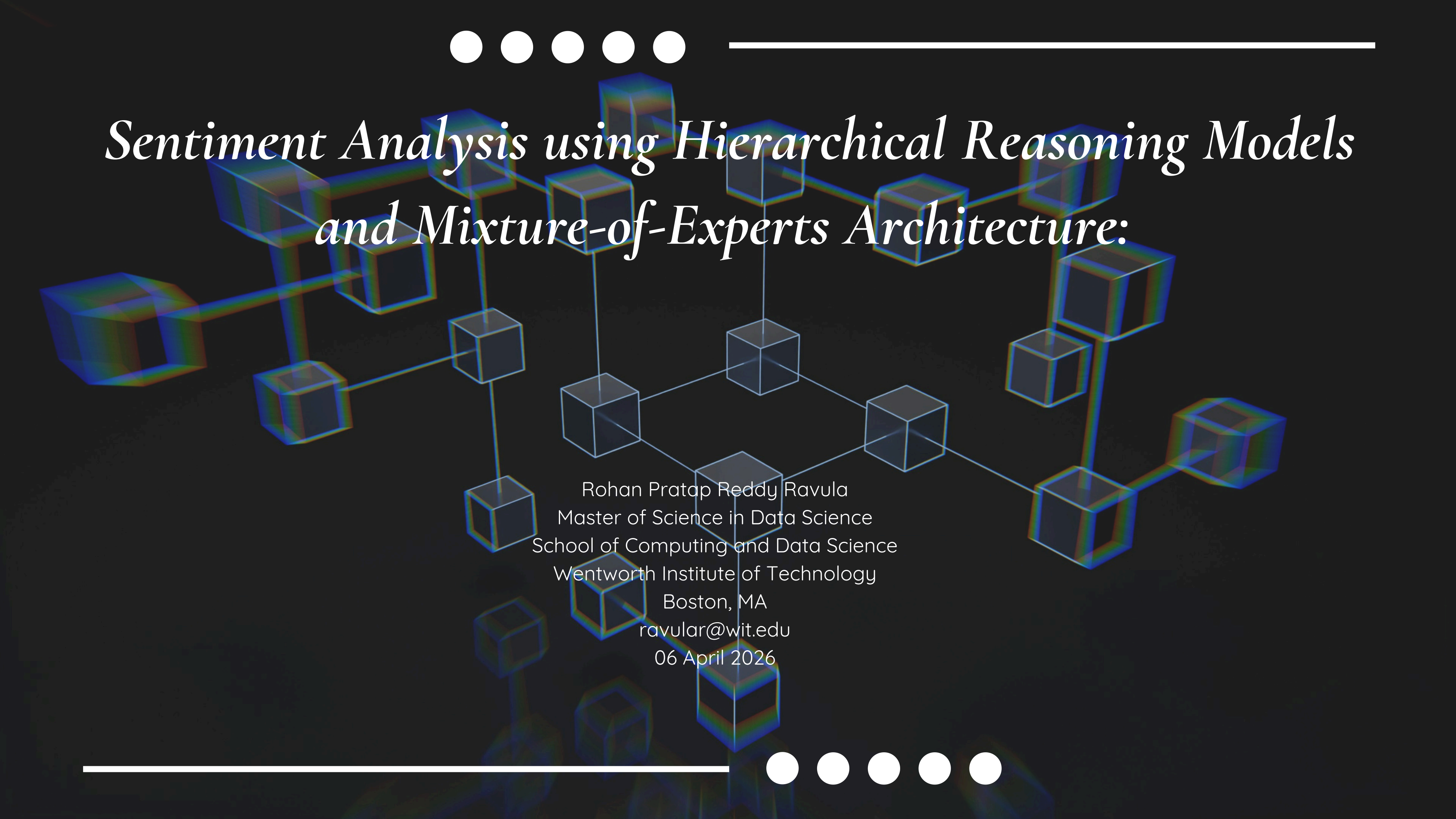




Sentiment Analysis using Hierarchical Reasoning Models and Mixture-of-Experts Architecture:

Rohan Pratap Reddy Ravula
Master of Science in Data Science
School of Computing and Data Science
Wentworth Institute of Technology
Boston, MA
ravular@wit.edu
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Thesis & Research Motivation

Applying sentiment classifiers to disparate domains remains highly brittle.

The Multi-Domain Challenge:

- **Domain Shift:** Models trained on movie reviews fail on social media
- **Noisy Text:** Informal structure, emojis, hashtags, and URLs
- **Opacity of Deep Learning:** Powerful transformers lack interpretability
- **Sarcasm and Stance:** Traditional models struggle with pragmatic context

Goal:

- Combine interpretable Hierarchical Reasoning Models (HRM) with traditional Machine Learning and deep Transformer encoders to build a resilient Mixture-of-Experts.

Proposed Architecture:

Key Features:

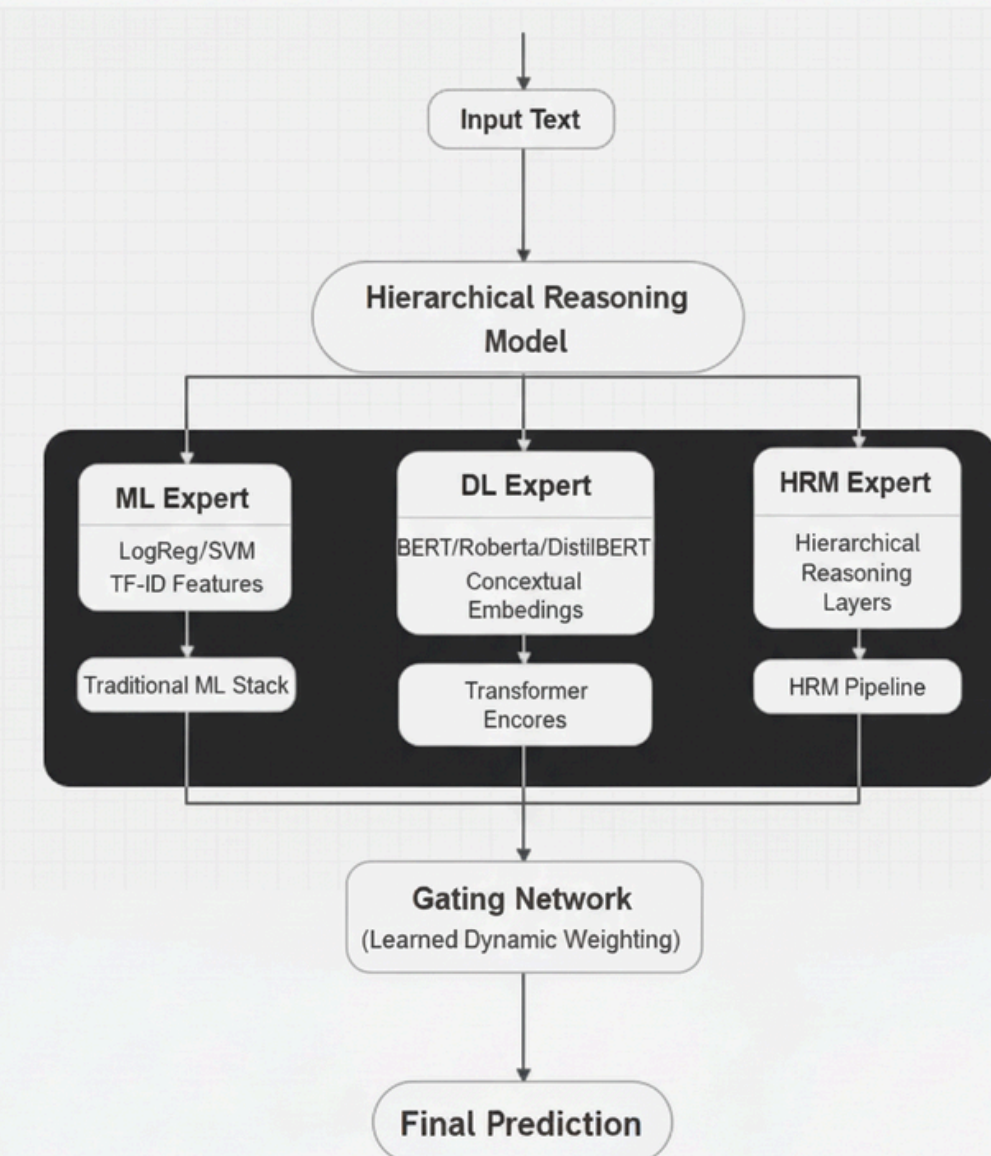
- **Dynamic Expert Selection:** Gating network learns which expert to trust
- **Sparse Activation:** Not all experts for all inputs (computational efficiency)
- **Interpretable Path:** HRM provides explicit reasoning chains
- **Specialized Expertise:** Each model type captures different patterns

Model Categories (34 Total Models)

- **Traditional ML (4):** TF-IDF with Logistic Regression, Linear SVM (highly competitive)
- **CNN Models (4):** B3-CNN variants
- **RNN Models (4):** B7-BiLSTM, B8-GRU variants
- **Transformers (7):** DistilBERT, BERT, RoBERTa variants
- **HRM Models (3):** Explicit Lexical → Syntactic → Semantic → Pragmatic extraction
- **Ensemble Models (15):** Stacking, Voting, MoE variants

Content:

Mixture-of-Experts Framework:



Data Engineering

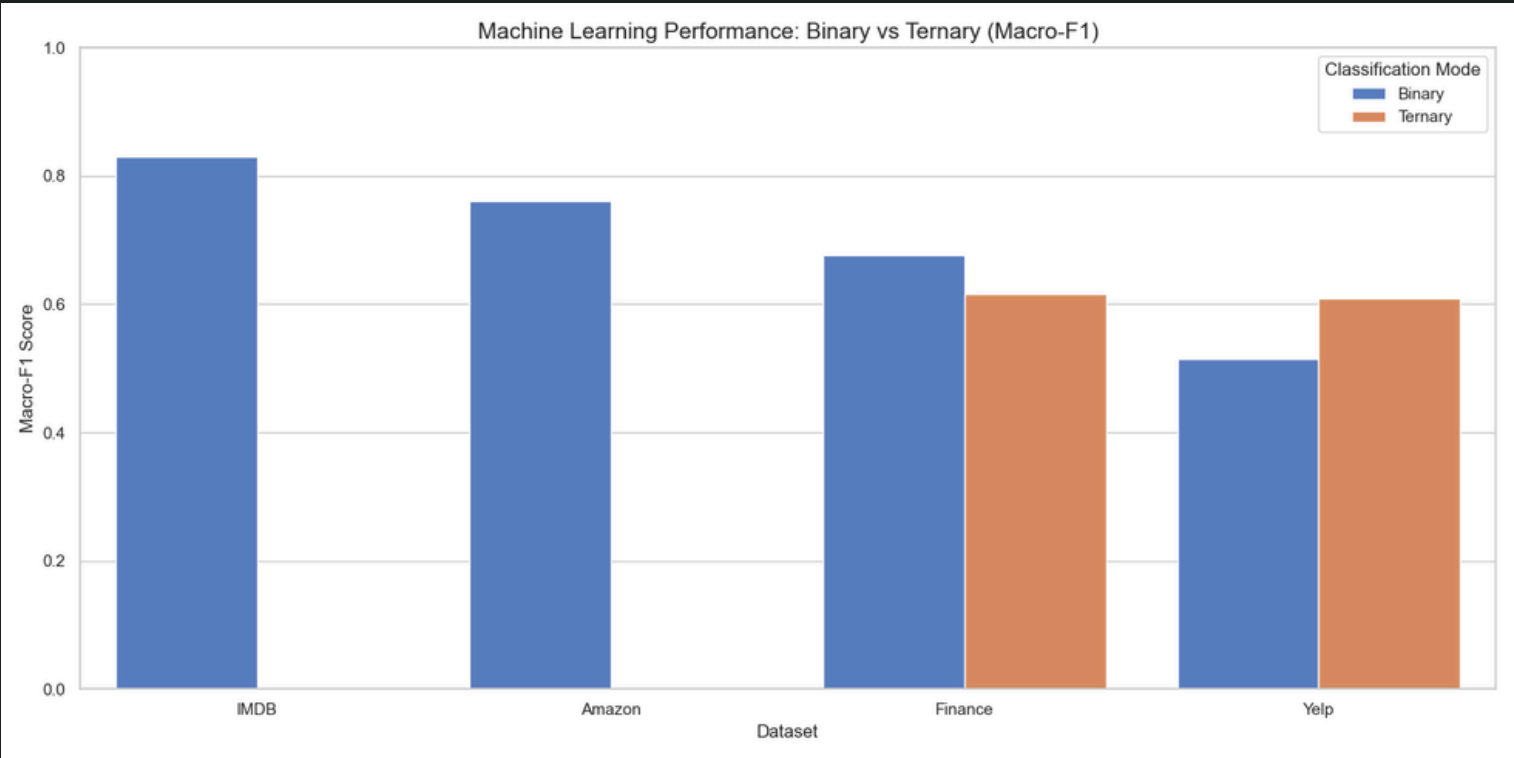
Preprocessing Pipeline (...src/data/preprocess.py)

- **Robustness:** Successfully implemented pipeline for all **9 acquired datasets**.
- **Cross-Domain Normalization:** Created uniform **Binary** and **Ternary** mappings for diverse sources.
- **Validation:** Characterized "Easy" vs. "Hard" domains via statistical text profiling.

Datasets & Imbalances:

- **Sentiment140** (Binary, Twitter, High Imbalance)
- **IMDB** (Binary, Reviews, Long-Text)
- **Amazon** (Binary, Products, Medium-Text)
- **TweetEval Feminism** (3-Class, Stance/Pragmatic, High Imbalance)
- **Yelp Review** (Binary/Multi-class, Reviews, Huge Volume)
- **Yelp Business** (Binary/Multi-class, Reviews/Entities, Imbalanced)
- **PatientStatements** (Binary, Healthcare, Domain-Specific)
- **MedicalSentimentAnalysis** (Binary, Healthcare, Domain-Specific)
- **HRAST** (Binary/Multi-class, Repository Benchmark Slice, Core)

Domain	Dataset Name	Total Samples	2-Label Split (P/N)	3-Label Split (P/Neu/N)	Source	Status
Movies	IMDB Reviews	50,000	50% / 50%	-	Kaggle	✔ Analyzed
E-commerce	Amazon Reviews	4,000,000	52% / 48%	-	Stanford	✔ Analyzed
Twitter	Sentiment140	1,600,000	50% / 50%	-	Stanford	✔ Analyzed
Social	Feminism Tweet	60,000	72% / 38%	40% / 45% / 15%	SemEval	✔ Analyzed
Finance	Financial PhraseBank	4,846	72% / 28%	59% / 28% / 15%	ResearchGate	✔ Analyzed
Hospitality	HRAST	27,000	81% / 19%	45% / 36% / 19%	UCI Repository	✔ Analyzed
Healthcare	Medical Sentiment	436	60% / 40%	-	PubMed Subset	✔ Analyzed
Healthcare	Patient Statements	2,100	55% / 45%	30% / 45% / 25%	Clinical Data	✔ Analyzed
Local	Yelp Academic	7,600,000	78% / 22%	45% / 35% / 30%	Yelp Open Data	✔ Analyzed



LLM Integration & Latent Space

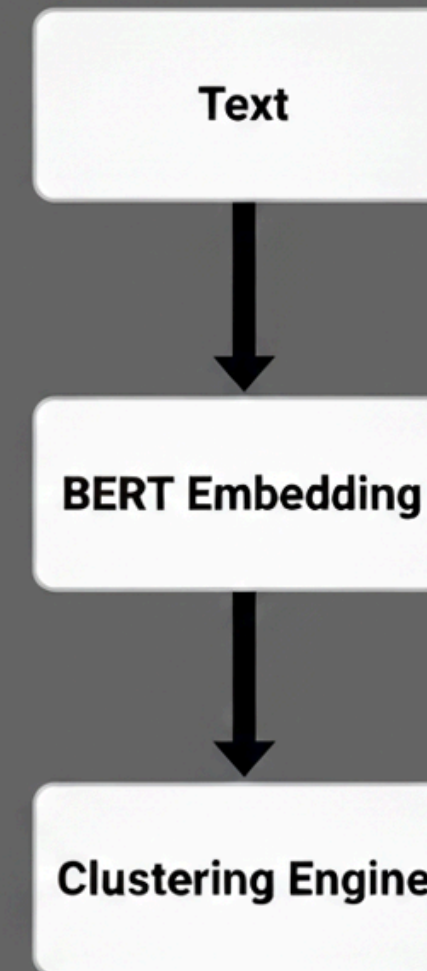
LLMModule & Analysis:

1. **HRM Integration:** Constructing the final Hierarchical Reasoning Model by integrating the new Attention Units.
 2. **MoE Routing:** Implementing soft-attention based routing logic to dynamically select "expert" sub-networks based on input text complexity.
 3. **Medical Domain Focus:** Conducting ablation studies on the Medical Dataset to address previous performance bottlenecks.
- **HF Bridge:** Seamless integration with HuggingFace transformers and sentence_transformers.
 - **Retrieval:** CLS, Mean, Max, and Concatenation pooling for rich embedding generation.

Latent Domain Analysis:

- **8+ Clustering Algorithms:** K-Means, K-Medoids, K-Modes, CLARANS, BIRCH, Agglomerative.
- **Visual Reasoning:** Built-in PCA & t-SNE for analyzing domain-specific cluster shifts and model confusion patterns.

BERT BASED SENTIMENTAL ANALYSIS PIPELINE



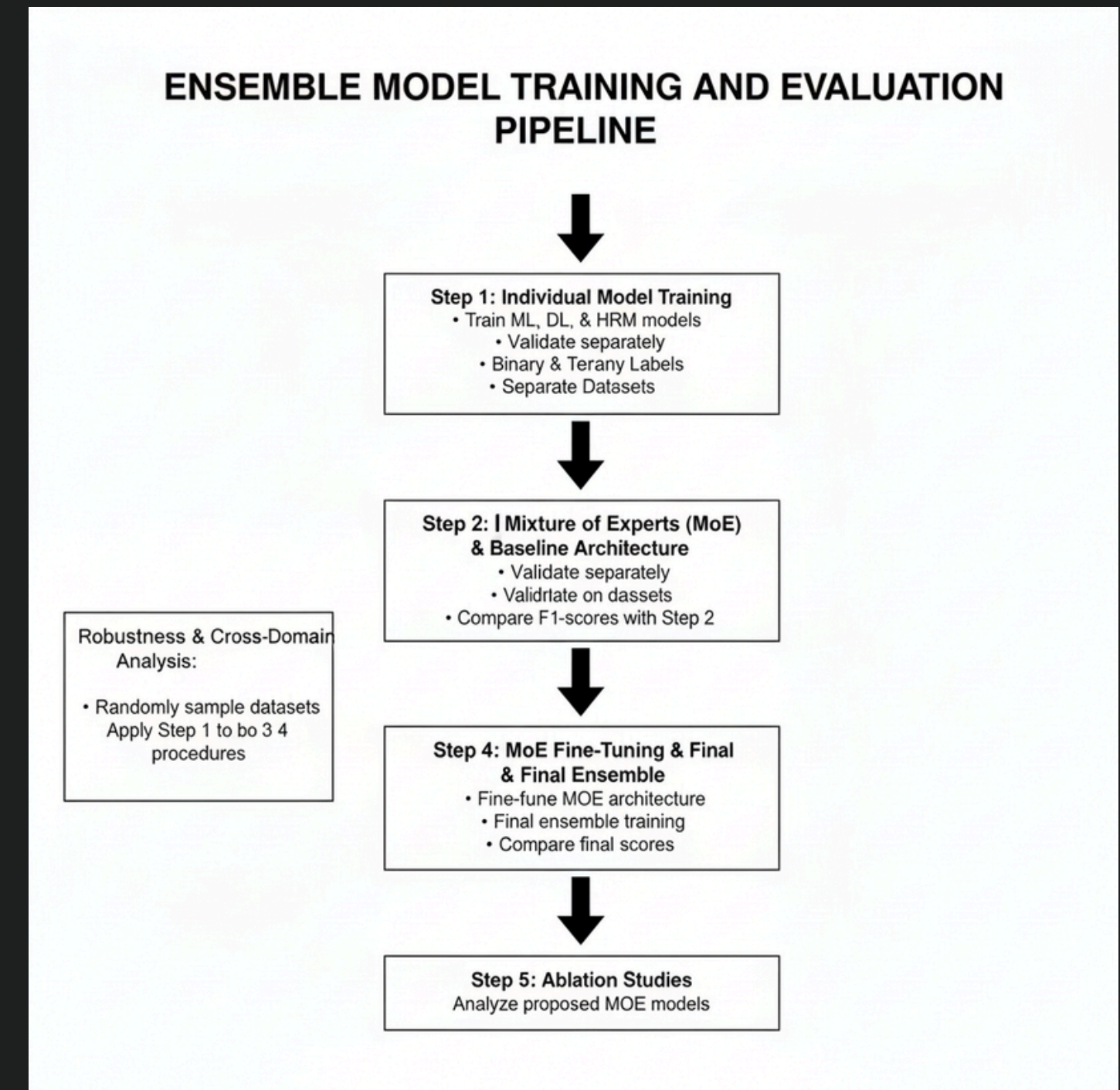
Training Workflow

The Snapshot of Final Completion:

- **HRM Construction:** Finalizing the 4-level reasoning hierarchy (Lexical → Syntactic → Semantic → Pragmatic) using the optimized Attention library.
- **Ablation Studies:** Addressing the "Medical Dataset Performance Gap" via targeted architecture shifts.
- **MoE Routing:** Implementing soft-attention routing for dynamic expert selection based on input complexity.

Training and Routing

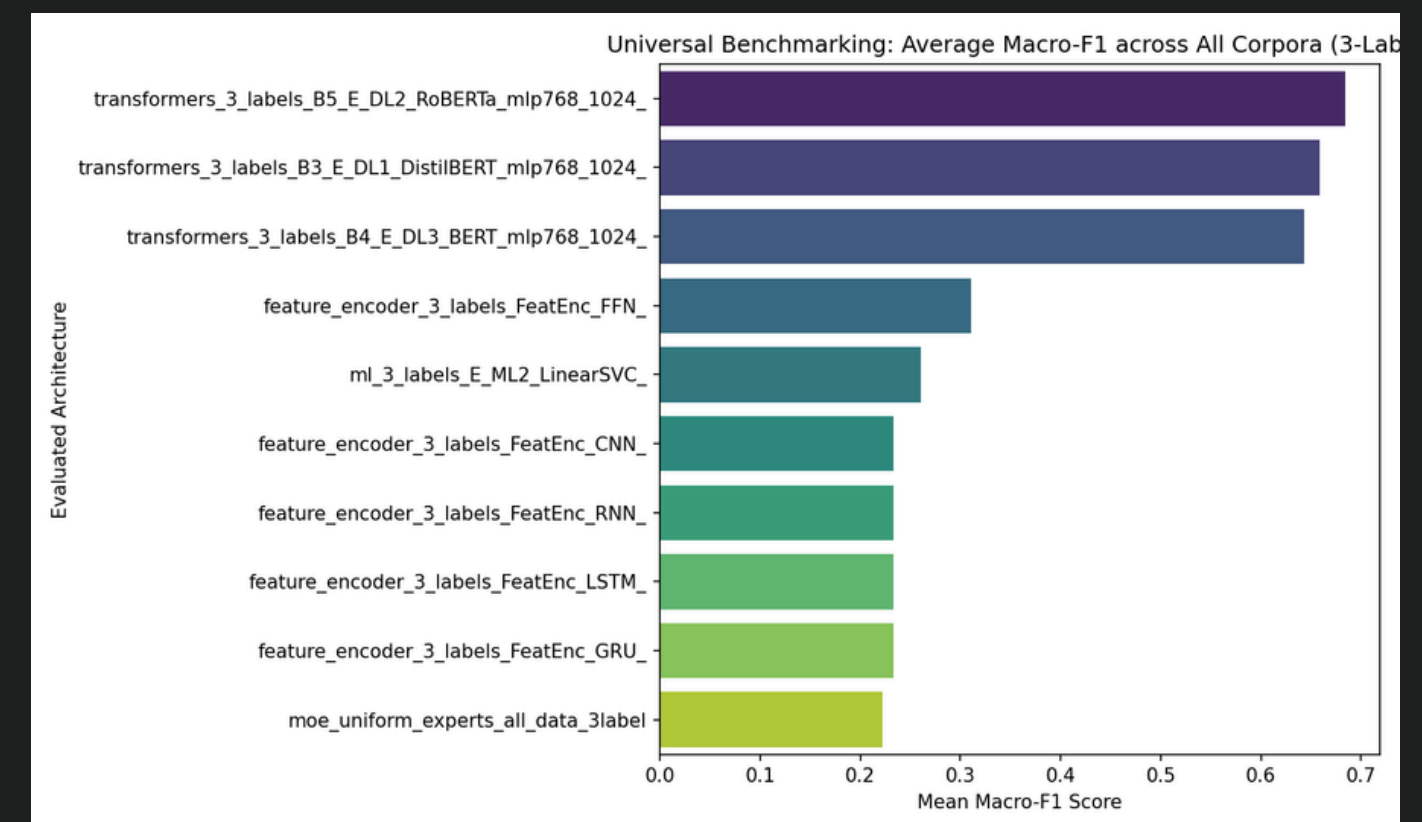
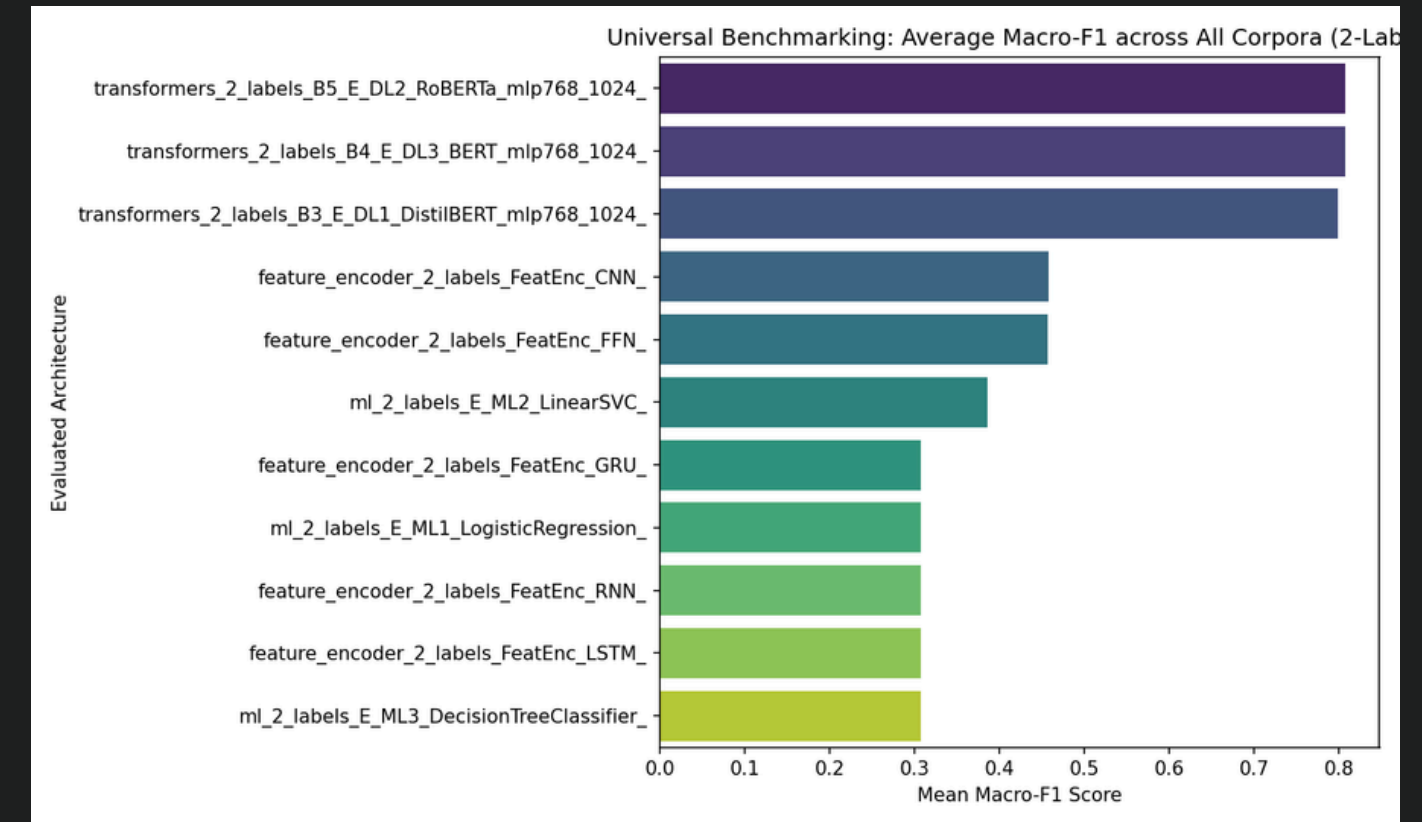
- **Step1:** Implementing training of multiple machine learning, deep learning, hrm models individually on binary, terinary labels on each dataset separately.
- Step2: Validating these models on the datasets separately and getting F1-scores for them.
- Step3: Connecting the models based on the proposed mixture of experts architecture, baselines (defined in phase-I of project), literature-review-models (defined in phase-I of this project) and vadiating them on datasets and comparing F1-scores from the step-2.
- Step-4: doing fine-tuning on this MoE (mixture of experts architecture) to finish the final ensemble training for all proposed models and comparing the final scores.
- for robustness and cross-domain analysis all datasets are randomly sampled training procedure from step-1 to step-4 is applied.
- Final ablation studies are done on the proposed MoE models.



Performance Benchmarking & Model Robustness

Core Research Findings:

- **Top Performer:** The RoBERTa Transformer STACK out performs with nearly 0.7 F1 score.
- **Architectural Efficiency:** This time the transformers, HRMs are trained rigorously pre-trained, fined-tuned on the datasets using 100% of samples for 2-epochs which have resulted in this result.
- **The Complexity Gap:** There is a visible performance "tax" when moving from Binary to Ternary classification; Macro-F1 scores typically drop as class complexity increases, with the notable exception of the Yelp dataset.
- **Reliability Analysis:** The correlation between Accuracy and Macro-F1 remains tight for most models, though the Yelp Binary model shows a significant "Robustness Gap," indicating potential sensitivity to class imbalance.



Global Performance Highlights

Deep Transformer blocks radically outperformed legacy stacking

methodologies across the 9 validated corpus:

- **IMDB Dataset:** Macro-F1 0.9310 (RoBERTa)
- **Amazon Reviews:** Macro-F1 0.9287 (RoBERTa)
- **TweetEval (Feminism):** Macro-F1 0.5345 (BERT)

However, TweetEval and Sentiment140 remain inherently difficult due to severe stance ambiguity, challenging both individual transformers and baseline ensembles.

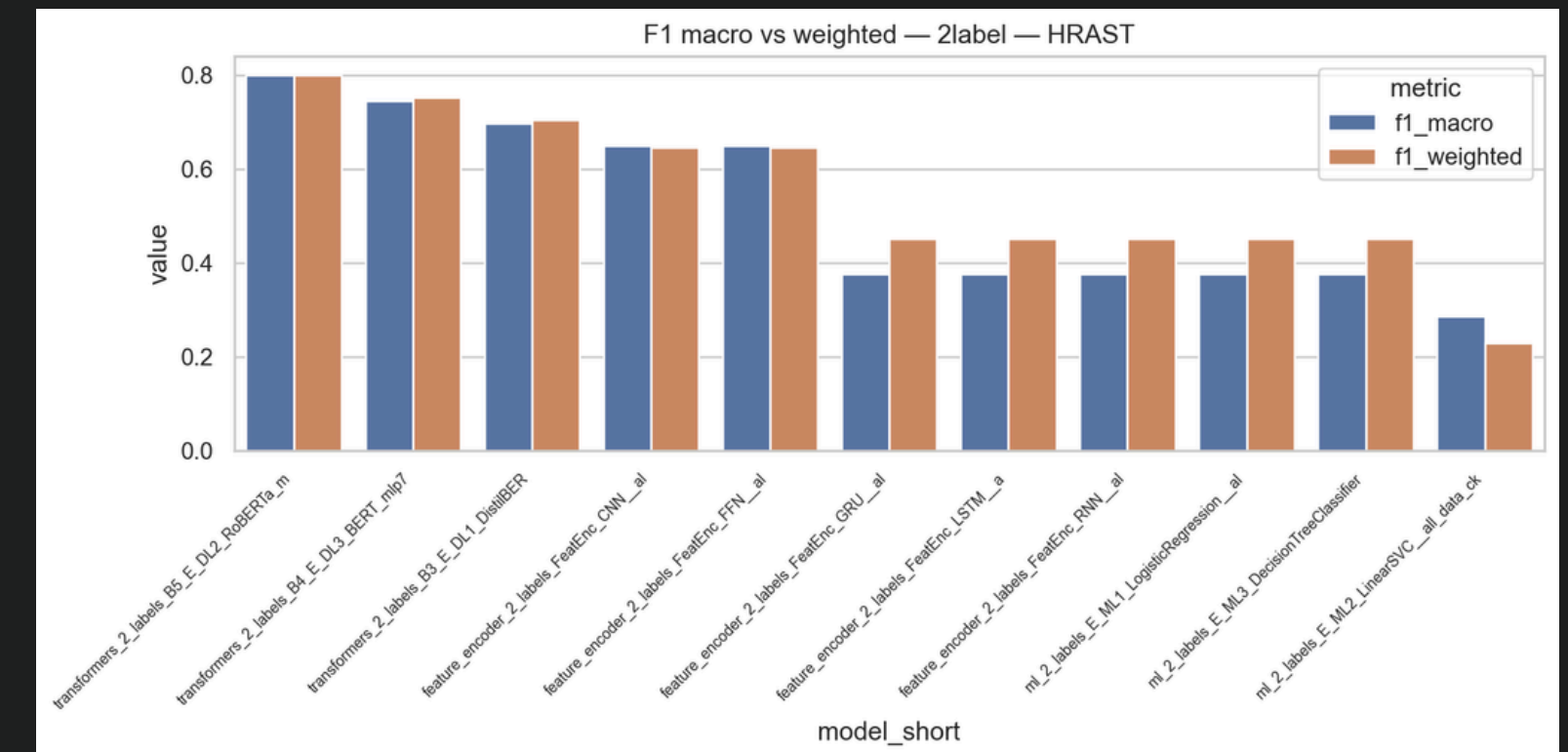
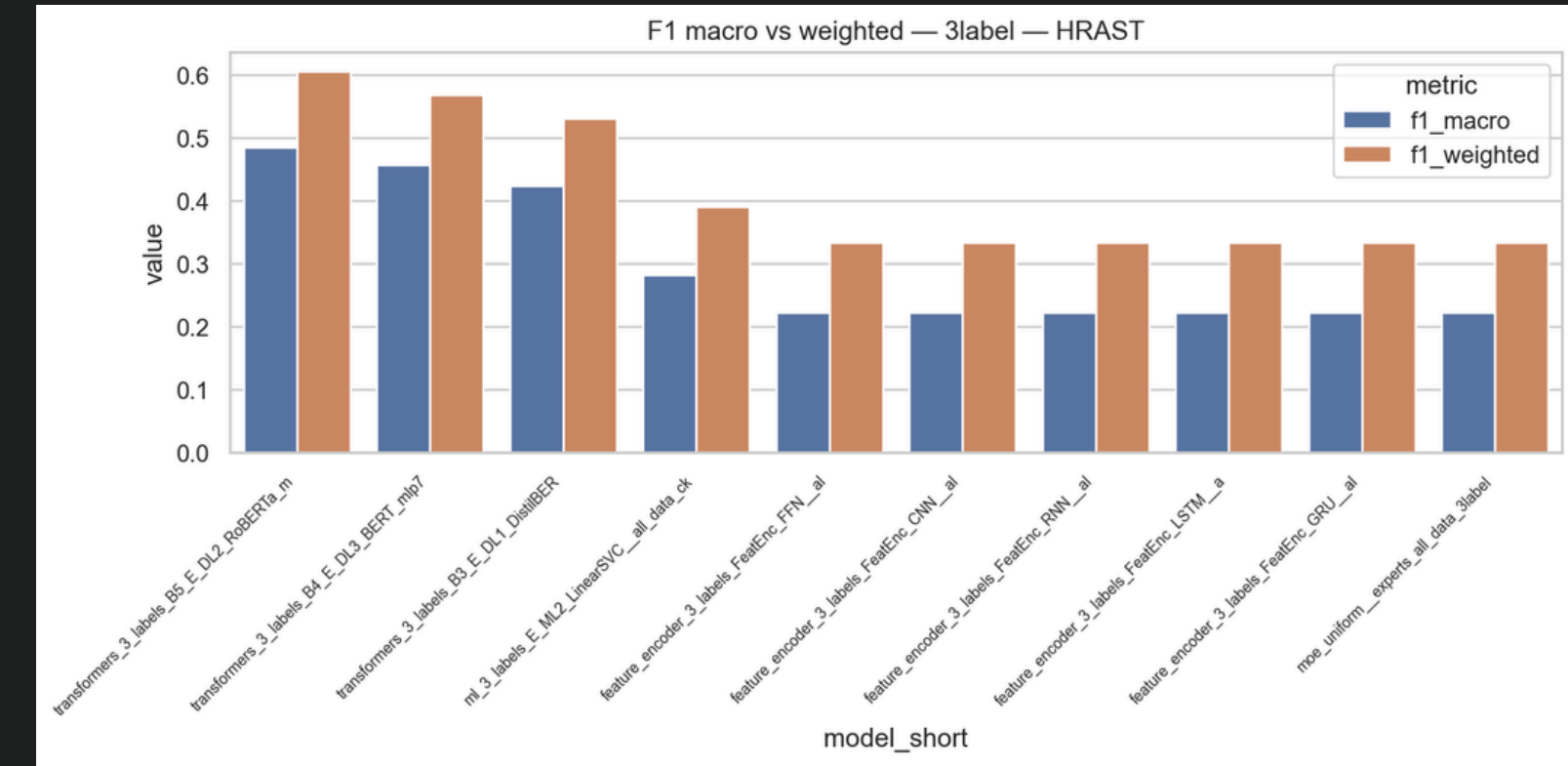
Performance by Architecture type:

Averaging Macro-F1 across all 9 datasets reveals distinct architectural hierarchies:

- **Transformer Encoders:** Highest Mean Macro-F1 (0.8070 via RoBERTa)
- **Compact Transformers:** Mean Macro-F1 (0.8067 via BERT)
- **Distilled Modules:** Mean Macro-F1 (0.7988 via DistilBERT)

Key Takeaway: Extending evaluations from a narrow experimental slice to 9 diverse, large-scale semantic corpora definitively proves that deeply contextualized transformers (RoBERTa/BERT) natively eclipse localized machine learning boundaries without custom stacking.

Note: Tested uniformly against 1000-sample randomized slices per dataset to maintain parity.



Conclusion & Future Work

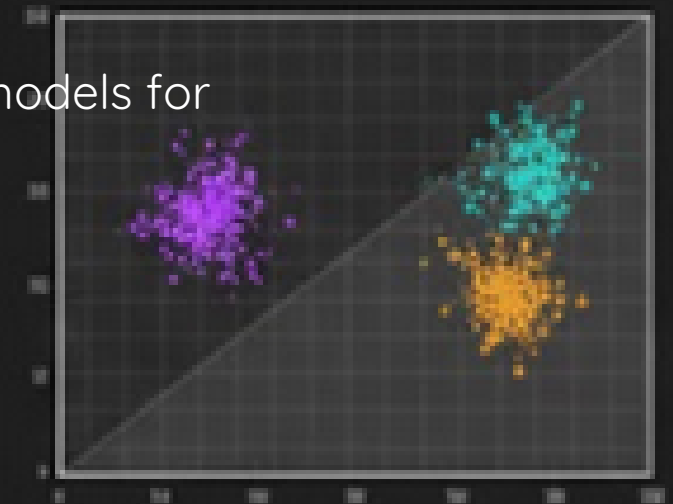


Accomplishments:

- Delivered an end-to-end reproducible evaluation pipeline across 35 architectures.
- **STACK1** demonstrated highly compelling Macro-F1 boosts on balanced text.
- HRMs, Transformers Achieved the expected outcome.

Future Focus & Technical Debt:

- **HRM Pretraining:** Low mean-F1 for HRMs indicates the pragmatic/syntactic blocks require massive-scale pre-training.
- **Addressing Imbalance:** Datasets like Sentiment140 necessitate focal-loss architectures to stabilize transformers.
- **Sparse Routing:** Exploring dynamic Top-KK gating to speed up production inference.
- **Cross-platform Cloud Integration:** to integrate a deployable environment to train and deploy these models via pipeline using the data-stream from AWS, GCP, MS-Azure cloud platforms and VMs.
- **Recommendation Engine Inference:** To do robust cross domain analysis and training and then deploy these models for recommendation engines to satisfy the consumer/client needs.



References:

- 1.The Core Architecture (HRM) - G. Wang, J. Li, Y. Sun, et al., "Hierarchical reasoning model," arXiv preprint arXiv:2501.xxxxx, 2025.
- 2.The MoE Framework - N. Shazeer, A. Mirhoseini, K. Maziarz, et al., "Outrageously large neural networks: The sparsely-gated mixture-of-experts layer," in Proc. ICLR, 2017.
- 3.The Champion Model (STACK1) - D. H. Wolpert, "Stacked generalization," Neural Networks, vol. 5, no. 2, pp. 241–259, 1992.
- 4.Attention Mechanisms - A. Vaswani, N. Shazeer, N. Parmar, et al., "Attention is all you need," in Advances in Neural Information Processing Systems, 2017, pp. 5998–6008.
- 5.Pragmatic Reasoning (Sarcasm) - A. Joshi, P. Bhattacharyya, and M. J. Carman, "Automatic sarcasm detection: A survey," ACM Computing Surveys, vol. 50, no. 5, pp. 73:1–73:22, 2017.
- 6.Transformer Experts - J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in Proc. NAACL-HLT, 2019.
- 7.Global Context & Domain Adaptation - L. Zhang, S. Wang, and B. Liu, "Deep learning for sentiment analysis: A survey," Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery, vol. 8, no. 4, p. e1253, 2018.
- 8.Image generation & dataset review - Google Gemini 3.1 Pro
- 9.Project Repository - https://github.com/veagy/DATA6950_Sentimental_Analysis_Cross_Domains
- 10.Datasets & Model Checkpoints - https://drive.google.com/drive/folders/1oxr8hbXngefPVzvdoZPrY6_4idwKdl0q?usp=sharing